

N+ - Network Fundamentals and building network

Network Cable & Connectors

[Q-1] Draw a labelled diagram of a Cat6 Ethernet cable, showing the colour order of wires in the T568B standard and labelling the RJ-45 connector pins.

RJ-45 Connector (Front View, Clip Away)

Pin: 1 2 3 4 5 6 7 8

Wire: W/O O W/G B W/B G W/Br Br

Legend:

W/O = White/Orange

O = Orange

W/G = White/Green

B = Blue

W/B = White/Blue

G = Green

W/Br = White/Brown

Br = Brown

[Q-2] List three types of network cables (such as Cat5e, Cat6, and fibre optic) and for each, mention one real-world app or device (like a WIFI router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Cable Type	Device/Application	Why Used
Cat5e	WIFI Router	Supports up to 1 Gbps speed and is affordable
Cat6	Zomato POS Terminal	Faster data transfer and less interference than Cat5e
Fiber Optic	IPL Streaming Server	Very high speed and long-distance data transmission

[Q-3] Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Cut the network cable to the required length.

Remove about 1 inch of the outer cable jacket.

Untwist and arrange the wires in the correct colour order (T568B is common).

Straighten the wires and trim them evenly.

Insert the wires into the RJ-45 connector.

Check that all wires reach the end of the connector.

Place the connector into the crimping tool.

Press the crimping tool firmly to secure the connector.

Test the cable using a cable tester.

[Q-4] Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Cable: Cat6 Ethernet Cable

Connector: RJ-45 Connector

Reason:

Supports high-speed internet for gaming and IPL streaming.

Reliable and stable connection with low lag.

Easy to connect multiple screens, consoles, switches, and routers.

[Q-5] Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WIFI router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

Example Description:

Type: Cat6 Ethernet Cable with RJ-45 Connector

Location: Behind a WIFI router

Connected Devices: WIFI router and broadband modem

Purpose: Provides internet connectivity from the modem to the router.

Sample Caption:

This is a Cat6 Ethernet cable with an RJ-45 connector. It connects a broadband modem to a WIFI router and provides a stable internet connection.

